DAY 28 | POST-READ

Protocols and Advanced Agents: Recap and Beyond

What we covered today, plus ideas to stretch your understanding.

Prepared by **Ahmed** | linkedin.com/in/ahmed-al-imaad

Part A: Summary of today

Topic	What to remember
Data analysis agent	Generate code, check the AST, run in a sandbox, show code and result.
Sandboxing (NEW v4)	Container per run, no network, read-only data, CPU, memory and time limits.
MCP	Open protocol connecting AI applications to tools, resources and prompts.
MCP 2026-07-28 (NEW v4)	Stateless requests, OAuth 2.1 for remote servers, Tasks, caching hints.
Current SDK (NEW v4)	mcp 2.x: MCPServer replaces FastMCP; mcp.Client connects in-process, stdio or HTTP.
A2A	Agent Cards, tasks, messages, artifacts. Complements MCP.
A2A v1.0 (NEW v4)	Signed Agent Cards: verify before you delegate.
Agentic AI Foundation (NEW v4)	Linux Foundation home for open agent standards such as MCP and AGENTS.md.
Handoff payload	Summary, facts, steps tried, reason, suggested next step, urgency.

Part B: Going beyond today

1. Migrating v1 MCP code

Old tutorials import FastMCP from mcp.server.fastmcp and pin mcp<2. On mcp 2.x that import fails with a message pointing to the migration guide. Rename to MCPServer, use mcp.Client, rerun your tests, then remove the pin. Ask Codex to do it in small steps, and review every diff.

2. Securing remote MCP servers

Use short-lived OAuth tokens with narrow scopes, check that each token was issued for your server, and log every call. Treat tool results as untrusted input: a malicious server can try to inject instructions.

3. Tasks for slow tools

Exports, scans and reports can take minutes. Return a task, let the client poll, allow cancel, and expire finished tasks after a TTL.

4. Agent registries

Enterprises keep a registry of approved agents, their signed cards and public keys, like an API catalogue. Delegate only to agents in the registry.

5. Managed code execution

Providers offer hosted code interpreters that run model-written code in isolated containers. For customer work, prefer these or your own container sandbox over in-process execution.

Part C: Check yourself

- What problem does MCP solve?
- Name three changes in the MCP 2026-07-28 specification.
- Why should a client verify a signed Agent Card?
- List five limits for a code sandbox.
- What changed for FastMCP in mcp 2.x?

Part D: Practice ideas for tonight

- Add a search_faq tool and an apology prompt to the MCP server.
- Tamper with a signed Agent Card and confirm verification fails.
- Run the data agent in Docker with --network none and compare with the subprocess sandbox.

Coming up on Day 30: LLM security and guardrails: OWASP Top 10 for LLM and agentic applications, indirect injection, a red-team suite and vulnerability report v3.

Yoi shuumatsu wo (良い週末を): have a great weekend